

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. in CSE Final Examination, 2022
MATH-2105: Linear Algebra

Total Marks: 70

Time: 3 Hours

Answer any 5 (Five) of the following Questions:

1. a) If A and B are idempotent matrices, then A + B will be idempotent if and only if AB = BA = 0. 5

- b) Define rank of a matrix. Find the rank of the matrix $\begin{pmatrix} 6 & -2 & 0 & 4 \\ -2 & -1 & 3 & 4 \\ -1 & -1 & 6 & 10 \end{pmatrix}$ 5

- c) Prove that the following system of linear equations is inconsistent: 4

$$\begin{aligned} x_1 + 2x_2 - 3x_3 &= -1 \\ 5x_1 + 3x_2 - 4x_3 &= 2 \\ 3x_1 - x_2 + 2x_3 &= 7 \end{aligned}$$

2. a) Define matrix polynomial. Find the Eigen values and corresponding Eigen vectors of the following matrix $A = \begin{pmatrix} 2 & 1 & 0 \\ 3 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}$ 7

- b) If $A = \begin{pmatrix} 2 & 3 & 4 \\ 4 & 3 & 2 \\ 2 & 4 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 1 \end{pmatrix}$, Show that $(A + B)^2 \neq A^2 + 2AB + B^2$ 7

3. a) What is called basis and dimension of a vector space? 1+1

- b) Does every matrix have an inverse matrix, Justify? If 1+4

$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$ then find the inverse of A.

- c) If U and W be two subspaces of $\mathbf{R}^4$ generated by the set of vectors $\{(1,2,1,1), (1,2,-1,2)\}$ and $\{(1,2,3,0), (2,2,2,2)\}$. Then find (i)dim(U+W) and (ii)dim(U ∩ W) 7

4. a) What is rank of the matrix? Find the rank of the matrix: 1+3

$$\begin{pmatrix} 1 & 2 & -3 & -2 & -3 \\ 1 & 3 & -2 & 0 & -4 \\ 3 & 8 & -7 & -2 & -11 \\ 2 & 1 & -9 & 10 & -3 \end{pmatrix}$$

4
1
4

- b) Define minors and cofactors. Prove that,

1+1
+4

$$\begin{vmatrix} 1 & a & a^2 & 0 \\ 0 & 1 & a & a^2 \\ a^2 & 0 & 1 & a \\ a & a^2 & 0 & 1 \end{vmatrix} = 1 + a^4 + a^8$$

- c) Prove that, $A^2 - 5A + 7I_2 = 0$ where $A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$

4

5. a) State and prove the Cayley-Hamilton theorem.

2+4

- b) What is characteristic matrix? Find the characteristic roots and associated Eigen vectors of the matrix

2+2
+4

$$A = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{pmatrix}$$

- c) If A and B are orthogonal matrices each of order n then the matrices AB and BA are also orthogonal.

4

6. a) For which value of λ will be the vector $v = (1, \lambda, 5)$ in $\mathbb{R}^3$ is a linear combination of the vectors

5

$$v_1 = (1, -3, 2) \text{ and } v_2 = (2, -1, 1).$$

- b) Show that the vectors $(1, 1, -1)$, $(1, 2, 3)$ and $(4, 5, -3)$ in $\mathbb{R}^3$ are linearly independent.

3

- c) Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear operator defined by $T(x, y, z) = (x + 2y, y - z, x + 2z)$. Find the rank and nullity of T.

6

7. a) Let $T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be the linear transformation defined by

8

$$T(x, y, z, t) = (x - y + z + t, x = 2z - t, x + y + 3z - 3t)$$

Find a basis and the dimension of the (i) Range space of T (II) Null space of T.

- b) Define diagonal matrix. Diagonalize the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ and find an invertible 2×2 matrix such that $P^{-1}AP$ is a diagonal matrix with the eigenvalues of A down the diagonal.

6

-a
-2